A method for ranking scientific documents based on relevance to the concept ‘the effects of climate change on intangible cultural heritage’, identifying subsets of documents balancing semantic closeness with semantic proximity, and subjecting these subsets to both computer-aided and manual reviews

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The problem: incomplete literature reviews

How does climate change shape the ways in which intangible cultural heritage (hereafter, HERITAGE) practices are defined, regulated, and maintained? To date, only three literature reviews exist on climate change and HERITAGE (published in 2017 [1], 2021 [2], and 2025 [3]). They focus primarily on direct effects, and provide limited insights into both indirect effects and everyday adjustments in practices, even though some of the individual studies they analyse do contain relevant evidence. This evidence can be summarised as follows [4]: information on everyday adjustments in practices is scant, direct effects dominate in cryospheric regions, and indirect effects dominate elsewhere—to such an extent that, outside cryospheric regions, whether heritage is maintained as a dynamic cultural practice appears to depend less on climate change itself than on how its impacts are evaluated and how prevailing economic rationalities determine the types of adaptation actions adopted.

This overlooking of evidence available within the literature analysed in these reviews compounds a far more important shortcoming: the reviews are not comprehensive, in that the literature they analyse represents only a subset of the full corpus of scholarship addressing the effects of climate change on HERITAGE:

- The **2017** [1] and **2021** [2] **reviews** were designed to complement each other in terms of temporal coverage, together spanning the literature up to 2020. Both relied on keyword-based searches incorporating *a small number of broad concepts* (for example, 'cultural resources' and 'cultural heritage'), combined with variations of the phrase 'climate change'. Both used relatively lenient eligibility criteria regarding (i) the prominence of climate change relative to other structuring conditions, and (ii) the level of detail devoted to HERITAGE in comparison with built heritage. Using this research design, they retained 11 and 20 documents, respectively. Given these lenient inclusion criteria, in the 2021 review—the only one to report this information—just a minority of the documents categorised as addressing HERITAGE focused on it exclusively and in depth.
- The **2025 review** [3], which I authored, extends coverage to mid-2024. It employs a similar keyword-based search, relying on *a small number of broad concepts*. However, it applies substantially stricter eligibility criteria concerning both the prominence of climate change among other structuring conditions and the depth of information provided on reported HERITAGE cases. As a result, although all 31 documents included in the 2017 and 2021 reviews were retrieved in the search process, only one met the stricter eligibility criteria. To identify additional relevant studies, the 2025 review also employed citation chasing. This approach enabled the identification of eight additional documents published before the earlier reviews' cut-off dates—and missed by both reviews—as well as three further documents published after the 2021 review's cut-off date.
- **Exploratory searches** in Clarivate's *Web of Science* database retrieve a further set of highly relevant studies, including work on botany and ethnobiology [5-8], archaeology [9-11], social and cultural studies [12, 13], atmospheric and climate sciences [14, 15], and socio-ecological studies [16, 17], among other fields [18, 19]. These fifteen documents meet the 2025 review's strict eligibility criteria, thus signalling that they are likely to bear most directly on the core issue. However, the document-identification strategies used in the 2017, 2021 and 2025 reviews are not well suited to retrieving these fifteen documents: doing so would require a large number of narrow concepts (for example, 'performing arts', 'knowledge of nature', and 'oral expressions'), rather

than the small number of broad concepts utilised in these three reviews. Using *a large number of narrow keywords* instead, a search of Clarivate's *Web of Science* database retrieves the documents in the three published reviews, and the fifteen documents identified using exploratory searches—along with approximately 300,000 other records, among which, given this scale, further relevant documents are likely to be present, although their number is unknown.

Taken together, these observations suggest that the 2017, 2021 and 2025 reviews may not merely be incomplete by a small margin: they could plausibly be missing a substantial body of directly relevant work. However, the size of the shortfall is currently unknown, as is the extent to which an analysis of the full corpus of relevant scholarship might bring to light insights not previously captured—namely new insights into how climate change shapes the ways in which HERITAGE practices are defined, regulated, and maintained.

The solution: an innovative method for identifying and analysing all relevant documents

As described in the Appendix, natural language models are ideally suited to scanning hundreds of thousands of records extracted from scientific databases. By capitalising on these and other capabilities, the method sketched in this section enables an analysis of the literature on climate change and HERITAGE that has the potential to overcome the shortcomings of existing literature reviews. Notably, the same method could be applied to other scholarly areas where the potentially relevant body of literature is too large to be reviewed comprehensively using traditional methods.

This method can be broken down into two phases. Phase 1 focuses on identifying the complete corpus of documents relating to the effects of climate change on heritage and retrieving selected metadata for each document. Phase 2 describes how to analyse this corpus both manually, focusing on a core set of documents, and computationally, using a substantially larger subset.

The following paragraphs provide a step-by-step description of each phase. Although the description refers to research tools and facilities available at the University of Bergen, comparable support is available at most universities in the Nordic countries and beyond.

Phase 1: ranking by relevance

Phase 1 relies on natural language models. Using such models, one can achieve two critical outcomes: first, analysing a far larger number of documents than has previously been possible in this field, thereby significantly reducing the likelihood that relevant material is overlooked; and second, distinguishing a core body of key texts from a broader set of highly relevant ones by ranking documents from most to least relevant and grouping them into tiers of relevance.

Task 1.1 involves retrieving all records from Clarivate's *Web of Science* database that include—in the title, abstract, or keywords—any of several possible forms of the terms in the following list, which is based on UNESCO's definition of HERITAGE: 'language', 'artistic', 'ritual', 'festivity', 'knowledge of nature', and 'craft'. Retrieval is available at no cost through NIB, the *National Bibliometric Infrastructure*.

Task 1.2 entails representing each abstract retrieved (approximately 300,000) through a vector with several hundred dimensions. This vector captures the abstract's semantic content—its meaning—in numerical form. Additionally, a comparable vector must be generated for a short text describing the core analytical focus of the exercise—namely, understanding the ways in which climate change affects how HERITAGE practices are defined, regulated, and maintained. Vectorisation can be performed using a tried-and-tested natural language model such as *e5-large* or *ClimateBERT*, which can be run on a standard workstation.

Task 1.3 determines how closely each abstract aligns with our core analytical focus, thereby allowing all abstracts to be ranked from most to least relevant relative to that core focus. This can be done using a well-established vector-space similarity measure such as *cosine similarity*. At the anticipated scale, this calculation can be carried out on a standard workstation. Should the number of records substantially exceed 300,000, however, the use of high-performance computing would be advisable. In such a case, through its participation in NRIS, the *Norwegian Research Infrastructure Services*, the University of Bergen has free access to super-computing resources.

Using the ranked list of abstracts, Task 1.4 involves defining two subsets of documents. The first (core) subset comprises the up to one hundred documents closest to the core analytical focus of this exercise. The second (extended) subset comprises these same documents together with additional lower-ranked but still highly relevant documents, up to a total of five hundred documents. These two tentative upper limits are based on exploratory searches. The exact number of documents in each subset can be determined using well-established distribution-threshold methods, such as the *elbow method* or *maximum margin detection*. The resulting similarity scores make it possible to define subsets of documents that balance proximity to the target representation—or *closeness*—and clear separation from less relevant documents—or *distinctness*.

Phase 2: automated and manual analysis

Phase 2 results in two analyses: one on the core subset and one on the extended subset. The core subset is analysed manually, whereas the extended subset is examined using computational text analysis methods.

Task 2.1 concerns the analysis of the core subset. It comprises two sub-tasks: identifying the principal onto-epistemological premises in the field and analysing all documents in the subset.

The effects of climate change on HERITAGE have been examined through diverse onto-epistemological premises—for example, nature–culture dualism versus relational ontologies, and objectivist versus constructivist understandings of climate change and responses to it. To avoid privileging any one premise, identifying the principal ones is key. This can be achieved through iterative engagement with key documents in the field, alongside online consultations with domain experts across disciplines. Once identified, the full range of onto-epistemological premises can be synthesised into a comparative analytical framework that: (i) shapes the analytical categories and dimensions through which the documents in the corpus are examined; (ii) guides the interpretation of heterogeneity, tension and contradiction across the corpus; and (iii) makes explicit the onto-epistemological assumptions underpinning the synthesis process.

The full text of each document in the core subset can be annotated using qualitative data analysis software such as NVivo, drawing on both inductive (data-driven) and deductive (framework-driven) coding. Deductive codes should cover at least six areas: direct impacts on HERITAGE practices, indirect

impacts on such practices, everyday adjustments in HERITAGE practice, climate change as an amplifying factor in relation to other structural conditions, resilience versus maladaptation in HERITAGE management, and relationships between built heritage and HERITAGE practices. Additionally, documents should be annotated for contextual parameters including, but not limited to, geographical location, climate change hazards involved, societal groups involved, HERITAGE practices involved, and prominence of climate change as a structuring condition. Regarding the latter, for each document it is important to contrast claims made in the document with evidence in the scientific and grey literature, to avoid over- or underplaying the role of climate change.

Task 2.2 involves preparing a summary of each document in the extended subset. Each summary could comprise three sections—introduction, findings, and conclusions—and could be approximately 1,250 words in length. To this end, the full text of all documents in the subset must be downloaded from NIB through a semi-automated process. On the basis of these full texts, rather than abstracts alone, a natural language model such as *LLaMA* can be used to generate summaries. The model can then assign a set of descriptive labels to each summary—at least one for all six deductive codes and all contextual parameters used in Task 2.1. For each descriptive label, the model can be asked to produce a short narrative annotation summarising the evidence supporting the assigned label.

Task 2.3 involves assessing the accuracy of the labels assigned by the model (see Task 2.2), to establish a final set of analytical dimensions for the extended subset analysis. To this end, for all documents in the core subset, model-assigned labels can be compared with those derived through manual coding (see Task 2.1). For each deductive code and contextual parameter, agreement between the two coding approaches can be assessed using a standard inter-rater reliability coefficient. Where agreement falls below an acceptable level, the corresponding code or parameter should be excluded from the automated analysis of the extended subset.

Task 2.4 involves analysing these labels and annotations using qualitative data analysis software. The analysis has two aims: to assess the relative prominence of the various analytical dimensions, thereby revealing patterns of dominance and variation across documents in the extended subset, where differences in order of magnitude—rather than precise frequency counts—are of primary interest; and to examine the amplifying role of climate change in relation to other structuring conditions.

Task 2.5 concerns the integration of the findings from the analyses of the core and extended subsets. While the manual review of the core subset delineates the current state of knowledge and identifies key knowledge gaps regarding the effects of climate change on HERITAGE, the analysis of the extended subset uncovers recurring patterns, mechanisms, and configurations in the literature. Integrating these analyses involves interpreting the conclusions of the manual review against the broader evidential context provided by the automated analysis, while exploring the scope and limitations of those conclusions.

Grounding the review of the core subset in disciplinary-neutral ontological and epistemological premises is essential, as it ensures that the analysis privileges no single disciplinary perspective. Subjectivity and positionality arising from those ontological and epistemological premises constitute the principal research ethics issue in this work. This issue manifests, for example, in decisions related to coding, the framing of findings, and the integration of different perspectives. To address the issue, particular care must be taken to document these decisions transparently.

Appendix: background on natural language models

Natural language processing (NLP) is a field of computer science concerned with the recognition, analysis, and manipulation of text and speech (Box 1). Put simply, natural language models analyse human language by converting phrases or words into numerical representations. These representations enable quantitative analyses, hence the appeal of these models outside computer science.

Numerical representations of human language take the form of multi-dimensional vectors. The process of creating these vectors—so-called vectorisation—is informed by patterns and relationships drawn from vast amounts of related text that the model uses as a reference. Establishing this reference is referred to as 'training'.

A wide range of NLP models can be employed in climate change adaptation research, ranging from small to large language models, and from specialised to general-purpose models. The primary differences between these models lie in the volume and diversity of data used during training.

Many of these models can perform tasks for which they were not explicitly designed. However, determining which model will perform best for a given task and dataset is not always straightforward—particularly in the absence of reference datasets or labelled data for benchmarking. Consequently, collaboration between domain experts and computer scientists is essential to test and identify suitable model candidates [20].

Box 1: Applications of supervised and unsupervised machine learning

> While the adoption of NLP methods and tools in research has grown significantly in recent years, the field itself can be traced back to the mid-1990s. Today, the most prominent approaches in NLP research are supervised and unsupervised machine learning—algorithms that learn patterns from data and generate outputs based on those patterns. By leveraging these learned patterns, NLP helps in understanding and generating text. For example, NLP is commonly applied in tasks such as topic modelling, summarisation, clustering, translation, and sentiment analysis.

In the context of climate change adaptation, NLP is increasingly used to analyse large corpora of texts, significantly reducing the human effort and time required to process and interpret scientific literature [20]. Notable applications include analysing the evolution, trends, and biases in adaptation literature across over 60,000 scientific articles [21]; reducing a corpus of just over 50,000 articles to 8,691, and mapping differences in adaptation actions [22]; and mapping climate impacts from over 100,000 studies [23]. The NLP models used in these projects are designed to identify patterns and semantic similarities in text, enabling statistical evaluation of whether two words or texts convey similar meanings. This capability allows researchers to efficiently process large corpora and identify articles that are similar to each other or relevant to a specific domain or description. These identified articles can then be further analysed using additional dimensions or metadata.

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